

Deploying app to cluster and Cluster members-84

1. Introduction to Application Deployment in a Cluster

- Clustering improves scalability, availability, and load balancing.
- When deploying applications in a clustered environment, the application must be accessible and synchronized across all cluster members.
- Example environments: WebLogic, JBoss/WildFly, WebSphere, IIS with ARR, Azure Kubernetes Service (AKS), Azure App Service Environment.

2. Steps for Deploying an Application in a Cluster

Step 1: Prepare the Application

- Package application (WAR, EAR, JAR, or container image).
- Ensure configuration files are cluster-aware (session persistence, database connection pooling, etc.).

Step 2: Access the Cluster Console

- Open application server admin console (e.g., WebLogic Admin Console, Azure Portal, Kubernetes Dashboard).
- Navigate to Deployments or Applications section.

Step 3: Select Cluster Target

- Instead of deploying to a single server, select Cluster as the deployment target.
- This ensures all cluster members receive the application.

Step 4: Configure Deployment

- Upload the application package.
- Configure:
 - Session replication
 - Load balancing
 - High availability failover
- Assign deployment plan if needed.

Step 5: Deploy

- Start deployment process.
- Application is distributed across all cluster members.
- Verify deployment logs for errors.

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Azure Kubernetes Service (AKS) manages your hosted Kubernetes environment, making it quick and easy to deploy and manage containerized applications without container orchestration expertise. It also eliminates the burden of ongoing operations and maintenance by provisioning, upgrading, and scaling resources on demand, without taking your applications offline. [Learn more](#)

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3. Verifying Deployment Across Cluster Members

- Access app via cluster load balancer endpoint (e.g., `http://cluster-lb:8080/myapp`).
- Check if requests are routed to different members.
- Stop one cluster member, check if the app is still available (failover test).
- Validate session persistence (e.g., login session continues on another node after failover).

4. Example: Azure-Based Cluster Deployment

- Azure Kubernetes Service (AKS)
 - Deploy application as container image.

- Create Deployment + Service.
 - Expose using LoadBalancer or Ingress.
 - Cluster members = pods across nodes.
- Azure App Service Environment
 - Deploy web app.
 - Enable scale-out to multiple instances (cluster members).
 - Azure Load Balancer distributes requests.

5. Cluster Member Management

- Cluster members are individual servers or instances that are part of the cluster.
- Admin can:
 - Add new members for scaling.
 - Remove unhealthy members.
 - Restart individual members without downtime.
- In WebLogic or WebSphere:
 - Cluster members are managed nodes/servers.
- In AKS:
 - Cluster members are pods on worker nodes.
- In Azure App Service:
 - Cluster members are scaled-out instances.

6. Key Benefits

- High availability: Application continues even if one server fails.
- Scalability: Add more members to handle increased traffic.
- Fault tolerance: Automatic failover to healthy members.
- Centralized management: One deployment action propagates to all nodes.

7. Practical Hands-on Tasks

1. Deploy a sample Hello World web app to a cluster (WebLogic, WildFly, Azure AKS, or Azure App Service).
2. Access the app via load balancer URL.
3. Stop one cluster member and verify if the app is still accessible.
4. Scale the cluster to add more members and confirm the app runs on new nodes.
5. Review cluster logs and deployment status.